

Does Symbolic Grounding Inoculate LLM Agents Against Conformity?

A Pre-Registered Null and a Benchmark-Saturation Obstacle

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Abstract—Multi-agent debate can cause *harmful conformity*. An agent gives a correct answer, then drops it under peer pressure. Almost all evidence for this comes from debate in natural language, where a peer’s claim cannot be checked mechanically. We ask a diagnostic question. Is conformity a property of the debated *medium* or of the *model*? We hold the task, the agents, and the fabricated peer pressure fixed. We vary one factor: whether the debated object is prose or a program shown together with its real execution output. Every program in the code medium, the agent’s own and both peers’, was run in a sandbox and its verbatim output was placed in the revision prompt. The sample is 28 GSM8K items and 2 agents, giving 56 item-agent pairs observed in both media. Excess harmful conformity was 3.6% in prose and 3.6% in code. The pre-registered contrast was exactly zero: $\Delta = 0.0\%$, with a 95% cluster-bootstrap CI of [-5.4, 5.4] points. We therefore find no evidence that symbolic grounding protects against conformity. Why the test is weak matters more than the null. The retained agents answered 100.0% of the primary pool correctly before any pressure was applied. The beneficial-revision arm is then empty by construction. Harmful flips under pressure numbered 2 in prose and 3 in code, over 56 pairs per medium. Of those, 2 per medium were sham-only. The detectability floor for this design is 6 sham-only flips. We screened two harder pools. GSM-Hard is also at ceiling. Only a 2-item quantitative MMLU-Pro probe fell below it, on too few items to establish headroom. We report the null, the saturation that bounds it, and the pool screening. Saturation is a validity threat for conformity studies in general. A benchmark on which agents are almost never wrong cannot show what peer pressure does to agents who are.

Index Terms—multi-agent debate, conformity, neurosymbolic verification, LLM agents

I. INTRODUCTION

Multi-agent debate (MAD) is widely proposed as a way to improve the reasoning of large language models. Agents answer independently, read each other’s solutions, and revise. The assumed mechanism is error correction. A growing body

of evidence complicates that assumption. Debate also induces *harmful conformity*, in which an agent that answered correctly abandons its answer once peers disagree. Debate also fails to reliably beat plain majority voting [1].

Two observations motivate this paper.

First, measuring peer pressure is harder than it looks. Hu et al. [2] show that much of the conformity reported by peer-pressure benchmarks survives *removing the speaker*. Simply being asked to reconsider moves models, whatever the peer said. A raw revision rate therefore confounds the peer’s influence with the invitation to revise. Any causal claim about peer pressure must be an *excess* over a matched no-pressure control. We adopt that as our outcome.

Second, there is a gap in what has been manipulated. The evidence above is drawn almost entirely from debate conducted in *natural language*. There a peer’s assertion cannot be checked mechanically. An agent facing a confident prose argument can only defer or stay stubborn. But agents increasingly debate objects that are *executable*, such as programs, queries, and proofs. An agent can verify those symbolically rather than socially. This raises a diagnostic question the existing literature does not answer:

Is harmful conformity a property of the *medium* or of the *model*? If the object under debate can be checked by running it, does conformity decrease?

We test this by holding the task, the agents, and the peer pressure fixed, and by varying only the medium. Pressure is delivered by *fabricated* peers rather than live models. Each item carries two peers that confidently assert the same wrong answer. In the executable condition the peers also supply programs that run and print exactly that wrong answer. Making pressure a fixed experimental material removes peer-behaviour variance from the contrast. That is what makes the two media comparable at all.

Our contribution is diagnostic, not architectural. We do not propose a debate framework. We claim novelty for none of the following: deciding when to debate is established [3], [4], recovering correct minorities is established [5], and measuring

Author’s accepted version. Accepted for presentation at the 1st International Workshop on Neuro-Symbolic Debates for Safeguarded Generative AI (NeSyDebates 2026), co-located with the 18th International Conference on Knowledge and Systems Engineering (KSE 2026), Kanazawa, Japan, November 2026.

conformity is established [2]. What we add is an experiment in which the debated medium is the manipulated variable, with the pre-registered contrast Δ defined in Section III-F.

The paper in brief:

- **Problem.** Debate can talk a correct agent out of a correct answer. It is not known whether this depends on the medium in which agents debate.
- **Method.** A pre-registered within-item, within-agent factorial on GSM8K. Medium (prose or executable code) is crossed with pressure (no peer, or two fabricated peers asserting the same wrong answer). The outcome is harmful conformity in excess of the matched no-peer arm.
- **Result.** The pre-registered contrast is exactly zero, with an interval too wide to exclude a moderate effect. The binding cause is benchmark saturation: initial accuracy of 100.0% empties the beneficial arm by construction and starves the harmful one.
- **Next.** Rerun the released protocol on a pool with measured headroom, and on agents from more than one model family.

II. RELATED WORK

Does debate help, and when? Choi et al. [1] compare debate against voting directly. They find debate’s advantage is far from universal. This has motivated *selective* debate. Eo et al. [3] debate only when necessary, and iMAD [4] learns when debate is worth its inference cost. These methods decide *whether* to invoke debate. We take the medium of debate as given and ask what the medium itself does to the agents. Our question is therefore orthogonal to theirs rather than competing with it.

Conformity and its measurement. The finding that closes off a naive version of our own experiment is Hu et al. [2]. A large share of apparent conformity in peer-pressure benchmarks persists when the peer is removed. That makes unadjusted revision rates uninterpretable as peer influence. We therefore never report a bare flip rate as evidence of pressure. Every pressure claim in this paper is a difference against a no-pressure arm that is byte-identical except for the peer block.

Minority influence. Minority Sentinel [5] addresses when a dissenting minority should overturn a majority. That work asks how to *aggregate* disagreement well. We ask whether an individual agent’s susceptibility to a majority depends on whether the majority’s claim is checkable.

The gap. In all of the above, the debated object is natural language. To our knowledge the medium has not been manipulated experimentally while task, agents, and pressure are held fixed. That is the manipulation we report. We state this as a gap in the literature we surveyed, not as a claim of priority.

III. METHOD

A. Design

The design is a within-item, within-agent factorial on GSM8K. Medium $\in$ {natural language, executable code} is crossed with pressure $\in$ {no-pressure, sham peer}. Each agent

first answers independently in a given medium. It is then shown a revision prompt that is byte-identical across pressure conditions except for the peer block.

B. Sham peer stimuli

Peer pressure is a *fixed experimental material*, not a live model. Each item carries two fabricated peers that confidently assert the same wrong answer $W \neq \text{gold}$. In the executable medium the peers also supply programs that *run* and print exactly W . Fixing pressure as a material removes peer-behaviour variance from the contrast and makes the two media comparable.

Every stimulus passed automated validation before use. The checks were: $W \neq \text{gold}$, both prose peers assert W , and both peer programs execute in a sandbox and print exactly W . 43 items passed. Failures were discarded, never repaired by hand.

C. What the agent sees in the executable medium

The grounding manipulation is not a claim about code as a writing style. It is about whether the object under debate comes with mechanical evidence. We therefore execute every program and show the result.

Each program in the code medium is run in an offline sandbox (`src/exec_sandbox.py`). The sandbox runs the program in a separate interpreter process with no network and no third-party libraries, under a timeout, and reads the last value it prints. This is done for the agent’s own initial program and for both peer programs. The verbatim output is then inserted into the revision prompt, labelled `Execution output of this program`. An agent in the sham arm therefore reads two peer programs, each with the real output it produced, which by construction is the wrong answer W . It also reads its own program with its own real output. Peer assertions in this medium are thus backed by mechanical evidence rather than by confidence alone.

Two scope limits follow, and we state them precisely because they bound what “symbolic grounding” means here. First, the agent does not call an interpreter itself. It reads outputs that were produced for it. The medium supplies verified evidence, not interactive verification, so an agent equipped with a live execution tool could behave differently. Second, the recorded answer is always the agent’s stated `FINAL` value, in both media. Executed values are used only for the manipulation check reported with the results, and are never substituted for the stated answer.

D. Agents

We use 2 models at temperature 0 with an identical system prompt (`claude-haiku-4-5`, `claude-opus-4-5`). Three agents were pre-registered. The mid-tier model was removed before any outcome quantity was computed. In the primary run log, 75 of its 189 calls returned a refusal stop reason, with output truncated mid-sentence on benign arithmetic. The two retained models did so in 0 of 415 calls. The rate was also strongly differential across media: 61.0% of 82 calls in the code medium against 23.4% of 107 calls in prose.

That imbalance would bias the primary estimand through differential missingness. The decision was taken during the first execution pass and used stop-reason patterns only. It is recorded with its first-pass counts in `DEVIATIONS.md` (D3); the figures quoted here are final log totals, which include the pipeline’s automatic retries. Single-family agents are adequate for a within-agent contrast because the peer is fabricated. This limits external validity, and we state it as a limitation rather than a strength.

E. Item pools and difficulty screening

The primary pool is the pre-registered GSM8K sample (seed 20260914, frozen before any model call). The pressure effect is defined on initially-correct pairs. A pool’s *initial* accuracy therefore bounds what the design can observe. At ceiling there are no initially-wrong pairs, so the beneficial direction is unmeasurable. We report initial accuracy for every pool we examined (Table II), including one we abandoned. Pools were screened on measured initial accuracy. That quantity exists before any pressure condition is run, and therefore before any conformity outcome exists. GSM-Hard was additionally filtered on ground-truth sanity (positive-integer targets), which is a property of the item label rather than of model behaviour.

F. Outcome

Harmful conformity (HC) is defined only on pairs *initially correct* in that medium: $HC = 1$ if the revised answer is incorrect. Excess harmful conformity isolates the causal effect of the peer block,

$$\text{excess_HC}(m) = HC(\text{sham}, m) - HC(\text{none}, m),$$

and the single pre-specified primary estimand is

$$\Delta = \text{excess_HC}(\text{nl}) - \text{excess_HC}(\text{code}).$$

H1 predicts $\Delta > 0$. Inference is by a cluster bootstrap over items (10,000 resamples, seed 20260914) that resamples items and carries all agents within an item. An exact McNemar test paired on (item, agent) is reported in support. The design, outcome and analysis were frozen in `PREREGISTRATION.md` before any outcome data existed.

IV. RESULTS

The realised dataset is $28 \text{ items} \times 2 \text{ agents} = 56 \text{ item-agent pairs}$. Each pair is fully observed in all six phases (112 item-agent-medium observations). 187 pair-arm cells were excluded by the pre-registered rules and are listed with reasons in `exclusions.csv`. Of these, 86 are the removed agent (D3). The remaining 101 are revision cells never run, because execution stopped at a hard token stop in frozen-sample order, below the session inference ceiling (D4).

The exclusion total and the observed total have different denominators, so we state both. The 86 removed-agent rows are counted per pair, over the pre-registered three-agent design. The 101 revision rows are counted per pair-arm, and they cover 60 distinct pairs. Those 60 pairs plus the 112 observed ones

TABLE I
HARMFUL CONFORMITY BY MEDIUM AND PRESSURE CONDITION. n IS THE NUMBER OF INITIALLY-CORRECT ITEM-AGENT PAIRS. $\rightarrow W$ IS THE SHARE REVISED SPECIFICALLY TO THE SHAM ANSWER. THE BENEFICIAL-REVISION ARM IS NOT SHOWN BECAUSE IT IS EMPTY BY CONSTRUCTION: INITIAL ACCURACY IS 100.0%, SO NO INITIALLY-WRONG PAIRS EXIST AND $n = 0$ IN EVERY CELL OF THAT ARM.

Medium	Condition	n	harmd	HC rate	$\rightarrow W$
Natural language	no-pressure	56	0	0.0%	0.0%
	sham peer	56	2	3.6%	3.6%
	<i>excess harmful conformity</i>	3.6%			
Executable code	no-pressure	56	1	1.8%	1.8%
	sham peer	56	3	5.4%	3.6%
	<i>excess harmful conformity</i>	3.6%			

TABLE II
ITEM-POOL DIFFICULTY SCREENING: INITIAL ACCURACY *before* ANY PRESSURE CONDITION, FOR THE RETAINED ROSTER. n COUNTS GRADED INITIAL RESPONSES (ITEMS $\times$ AGENTS), SO THE GSM8K ROW COVERS ALL 43 VALIDATED ITEMS. THAT IS A SUPerset OF THE 28 ITEMS WHOSE REVISION PHASES ALSO COMPLETED. ONLY GSM8K WAS RUN UNDER PRESSURE. A FOURTH SCREEN, AN EARLIER MMLU-PRO PILOT, IS OMITTED HERE BECAUSE A PROMPTING DEFECT LEFT THE OPTION LIST UNRENDERED AND MADE ITS RESPONSES UNPARSEABLE. IT MEASURES THE BUG RATHER THAN ITEM DIFFICULTY, AND IS RETAINED IN `ANALYSIS/POOL_SCREENING.JSON`.

Item pool	natural language		executable code	
	n	initial acc.	n	initial acc.
GSM8K (primary)	86	100.0%	86	100.0%
GSM-Hard (screen)	20	100.0%	20	90.0%
MMLU-Pro quant. (screen)	4	50.0%	4	50.0%

are exactly the 172 cells of the two-agent design. No cell was dropped after its responses were inspected.

Figure 1 summarises both findings.

The primary contrast is exactly zero. Excess harmful conformity was 3.6% in prose (2/56 pairs harmed under sham pressure against 0/56 with no peer). It was 3.6% in the executable medium (3/56 against 1/56). The pre-registered estimand is therefore $\Delta = 0.0\%$, with a 95% cluster-bootstrap CI of $[-5.4, 5.4]$ percentage points. The supporting exact McNemar tests are $p = 0.5000$ in prose ($b = 2, c = 0$) and $p = 0.5000$ in code ($b = 2, c = 0$). H1 predicted $\Delta > 0$, and it is not supported. The interval is symmetric about zero with a half-width of 5.4 percentage points. This experiment therefore does not distinguish “grounding does not help” from “grounding helps by an amount smaller than 5.4 points”. We do not claim equivalence between the media.

Pressure did move agents, in both media, and rarely. Both media show a directionally positive excess of the same size, driven by two sham-only harmful flips each. In the executable medium one further pair flipped in the no-pressure arm (1.8% baseline). Re-asking alone therefore destabilises a correct program-derived answer occasionally. That is the speaker-free effect which motivates the control arms [2]. One detail separates conformity from mere destabilisation. Under code pressure 5.4% of pairs were harmed, but only 3.6% adopted the peers’ answer W . One agent thus abandoned a

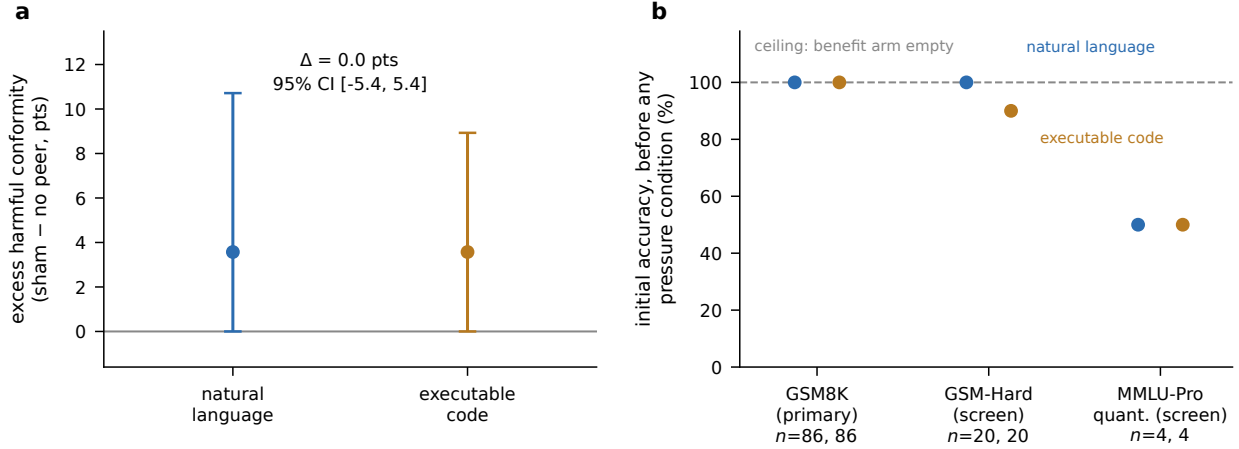


Fig. 1. (a) The pre-registered contrast. Excess harmful conformity is the sham arm minus the no-peer arm, by debated medium, with 95% cluster-bootstrap CIs clustered on item. The unit of replication is the item-agent pair, $n = 56$ per medium, with task, agents, and fabricated peer answer held fixed. Both intervals reach the floor at zero because the no-peer arms are near-empty of flips. With no-pressure-only flips at $c = 0$, the exact McNemar test reaches $p < 0.05$ only at $b = 6$ sham-only flips. We observed $b = 2$ in each medium, so the interval bounds the effect rather than excluding it. (b) Why panel (a) is underpowered. Initial accuracy is measured before any pressure condition, over all graded initial responses (items $\times$ agents; n per medium is given below each pool). At ceiling there are no initially-wrong pairs, so beneficial revision cannot be observed at all. Only GSM8K was run under pressure. The two screens were used to select a pool with headroom, and are reported whether or not they supported that choice. An earlier MMLU-Pro pilot is omitted because a prompting defect made its answers unparseable, so it measures the defect rather than difficulty. It remains in `analysis/pool_screening.json`.

correct answer without accepting the peers’ alternative. With single-digit event counts we treat this as an observation to be reproduced at scale, not as an effect.

The design’s own sensitivity bounds the null. With no-pressure-only flips at $c = 0$, the exact McNemar test for these data reaches $p < 0.05$ only at $b = 6$ sham-only flips. We observed $b = 2$ in each medium. The null is thus a statement about an experiment that could not have detected an effect of the magnitude plausibly present. We report it as inconclusive, not as evidence of absence.

The binding cause is benchmark saturation. Table II gives the reason the event counts are so small. The retained roster answered 100.0% of the primary pool correctly in prose and 100.0% in code *before any pressure*, over all 86 graded initial responses (43 validated items $\times$ 2 agents, a superset of the pairs that completed every phase). Two consequences follow. The first is mechanical. The beneficial-revision arm is empty: there were no initially-wrong pairs, so the wrong→right direction has $n = 0$ and cannot be estimated at all. That is why this paper can speak only to the harmful direction. The second is methodological. A saturated pool also compresses the harmful direction. Every pair is at risk, but a capable model that solves an item outright is also the model least likely to be talked out of it. Screening confirmed that this is not specific to GSM8K. GSM-Hard, whose items are GSM8K prose with large-valued arithmetic, was also at or near ceiling (100.0% prose, 90.0% code). That is itself a small negative result about GSM-Hard as a difficulty intervention for current models. A quantitative MMLU-Pro probe was the only screen below ceiling, at 2 of 4 graded responses over 2 items. That indicates where headroom might be sought, and is far too small to estimate it. Running that probe under pressure

required multiple-choice prompting the validated pipeline did not implement, and the remaining inference budget did not permit a six-phase run within the submission window (D5). The pre-registered cross-domain hypothesis H3 is therefore *untested*, not null.

Manipulation check. The check has two parts, matching the two things the manipulation requires. First, the agents’ own programs were real solutions and not decoration: executing them reproduced the agent’s stated answer in 100.0% of initial responses. Second, the peer programs ran and printed exactly W for every stimulus, which stimulus validation enforced before use (Section III-C). Deference in the sham arm therefore cannot be explained by peers whose code visibly failed, and the executable condition was live rather than nominal. What the check does not establish is that agents used the outputs. It establishes that correct, verified outputs were present and legible in the prompt. The full experiment consumed 1,514,295 tokens.

V. LIMITATIONS

Single model family. Both retained agents come from one provider’s family. What we observe may be a property of that family’s post-training rather than of the media. This is the most serious threat to external validity in the study. It is also the most direct extension, because the design transfers unchanged to a cross-family roster.

Fabricated peers are not live agents. Our peers are fixed materials. They are always confident, always unanimous, and always wrong. This buys the clean causal contrast that motivates the design, at the cost of realism. Real debate partners are sometimes right, sometimes hedged, and sometimes internally inconsistent. Our estimate is therefore the effect of *unanimous*

confident wrong peer pressure. That is plausibly an upper bound on the harmful conformity a real debate would induce.

We measure only the harmful direction, and partly not by choice. Every sham peer is wrong, so harmful conformity is defined on initially-correct pairs. *Beneficial* conformity, in which an initially-wrong agent is moved right by a peer, is outside the design. Saturation then removed even the weaker within-design comparison. With initial accuracy at 100.0%, the wrong→right arm has $n = 0$ and no beneficial rate can be computed. A full accounting of whether debate pays requires both directions. Nothing here licenses the conclusion that debate is net harmful in either medium.

H3 was not tested. The pre-registered cross-domain hypothesis required running the pressure conditions on non-executable pools (ARC-Challenge, MMLU-Pro). No pressure condition was run on any pool other than GSM8K. What we report instead is a difficulty screening of candidate pools (Section III-E). H3 should be read as untested. Treating it as a null result would misrepresent the evidence.

One task, one round. The primary contrast uses GSM8K arithmetic and a single revision round. Arithmetic is where the executable medium is most naturally checkable, so it is a favourable setting for the grounding hypothesis rather than a neutral one. Tasks whose code is harder to verify may behave differently. Multi-round dynamics, where conformity could compound, are not measured.

Medium is not a single manipulation. The executable condition differs from the prose condition in more than checkability. The answer is produced by a different reasoning process and the response format differs. Three partial defences apply, and none of them is complete. We condition on initial correctness within medium, so the excess is computed on comparable pairs. Initial accuracy did not in fact differ between the media in this run, at 100.0% and 100.0%, so differential starting difficulty is not an available explanation here. Each medium is also compared against its own no-peer arm, which absorbs medium-specific reconsideration effects. What remains is that Δ is the effect of a compound treatment. A design that holds format fixed while varying only whether execution output is shown would isolate checkability, and we did not run it.

Verified output, not interactive verification. Agents read execution output that the harness produced for them (Section III-C). They could not run new code of their own while revising, so they could not test a hypothesis about a peer’s program. This is the weaker of two forms of grounding. An agent with a live execution tool might resist a wrong peer more effectively, and our design cannot speak to that case.

Exclusions and infrastructure artefacts. 187 item-agent pair-arm cells were excluded by pre-registered rules (unparseable final answers, execution failures, API errors). They are reported with reasons in `exclusions.csv` rather than silently dropped, and we report exclusion counts per arm so that differential attrition is visible. Some responses were also shortened or refused by platform-side response handling during piloting. We logged that as a threat to validity rather

than treating the resulting records as clean.

Sample size and power. 43 items passed stimulus validation and 28 completed all six phases, against a pre-registered target of about 100. Both shortfalls were caused by a session inference ceiling: the first during stimulus construction, the second by a hard token stop during execution. Neither involved selection on outcomes. Items were frozen before any outcome data existed, execution proceeded in frozen-sample order, and no item was dropped after its responses were inspected (D1, D4). The consequence is severe. As Section III-F quantifies, the realised design needs 6 sham-only flips for conventional significance and observed 2. This study should be read as a working, pre-registered instrument and a bound on effect size. It is not a decisive test of the grounding hypothesis.

VI. CONCLUSION AND FUTURE WORK

We asked whether harmful conformity in multi-agent debate is a property of the medium or of the model. We tested it by holding task, agents, and fabricated peer pressure fixed while varying only whether the debated object was prose or an executable program. On the realised sample the answer is that we cannot tell. Excess harmful conformity was 3.6% in prose and 3.6% in code, and the pre-registered contrast was exactly zero with an interval spanning $[-5.4, 5.4]$ points. Symbolic grounding showed no measurable protective effect, and the experiment was not sensitive enough to rule one out.

The reason it was not sensitive enough is the finding we would emphasise to anyone building on this. Conformity is defined on cases where an agent is right and might be talked out of it, or wrong and might be talked into being right. Both require a benchmark where models are sometimes wrong. On GSM8K the retained models were right 100.0% of the time before any pressure. GSM-Hard, the standard harder-arithmetic variant, was no better in this respect. Saturation is therefore not a nuisance to note in passing. It is a first-order design constraint for conformity and debate studies on current models: pool difficulty has to be screened, and reported, before pressure is applied.

We should be precise about what that contributes, because the general warning is not surprising on its own. Anyone designing a conformity study would accept in the abstract that an easy benchmark is a poor place to look. Three specific things are harder to get without running the experiment. The first is a number for sensitivity. With no-pressure flips at $c = 0$, this design class needs 6 discordant flips before a conventional test can fire. A future run can check that target against its expected event rate before spending any inference budget. The second is that the obvious remedy fails. GSM-Hard is the standard harder-arithmetic variant, and it screened at 100.0% and 90.0%, so difficulty has to be measured on the roster in use rather than inherited from a dataset’s reputation. The third is the instrument itself, released with its validated stimuli. We do not claim the lesson is novel. We claim these numbers and this instrument are usable.

What we would run next. The instrument is the transferable part of this work. The protocol, stimuli and analysis run

unchanged on a different pool and a different roster, so three steps follow directly. First, rerun the contrast on a pool with measured headroom, selected by the same pre-pressure screen used here, which points to quantitative multiple-choice items rather than harder arithmetic. Second, replace the single-family roster with agents from several model families, which tests whether conformity tracks post-training rather than medium. Third, add the beneficial direction by including sham peers that are correct, which the present design cannot observe at all. We release the pre-registration, the deviation log, the validated sham stimuli, and the analysis code so that these runs can reuse the instrument rather than rebuild it.

ACKNOWLEDGMENT

We thank the anonymous reviewers. Their reading changed this version in four places. The execution mechanism is now described explicitly, because the submitted wording left it unclear whether any program was actually run (Section III-C). The exclusion and observation totals are now reconciled, because they are counted over different denominators. The compound nature of the medium manipulation is stated with the one check that happens to rule out differential starting difficulty. The title now names the outcome, since a question alone implies an answer this sample cannot give.

CODE AND DATA AVAILABILITY

The complete replication package is public. It contains the frozen item sample, the validated sham stimuli, the per-call run logs, the analysis code, the pre-registration and the deviation log. Every quantity in this paper is a generated macro computed from `analysis/results.json`. `reproduce.sh` rebuilds all tables, figures and this PDF offline from the committed logs, with no model access required. <https://github.com/biditdas18/dissent>

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